Materie → Regim de curent continuu (toate sunt constante în timp)

+ Liniare

Regim de curent alternativ (sinusoidal)

+ Liniare

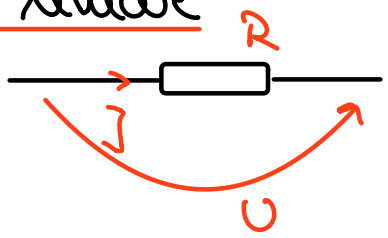
Regim nesinusoidal

Regim tranzitoriu

Circuite electrice de curent continuu

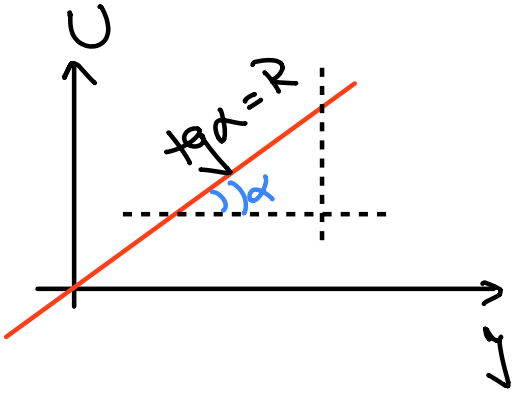
Elemente de circuit

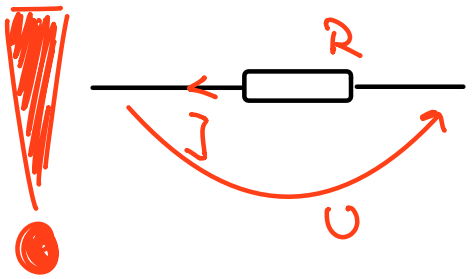
(1) Rezistorul liniar



rezistență (R)

$$U = R \cdot I$$





$U = -RI$

$$I = G \cdot U$$

$$G = \frac{1}{R}$$

conductanță

$[G]_{SI} = 1S$ (Siemens)

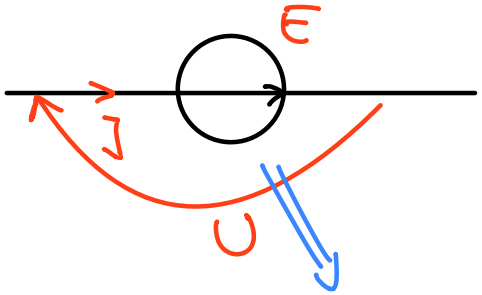
Cazuri particulare

$R = 0;\ G \to \infty$ → $U = 0$ scurtcircuit

$G = 0;$

$$P = U \cdot I = RI^2 \geq 0$$

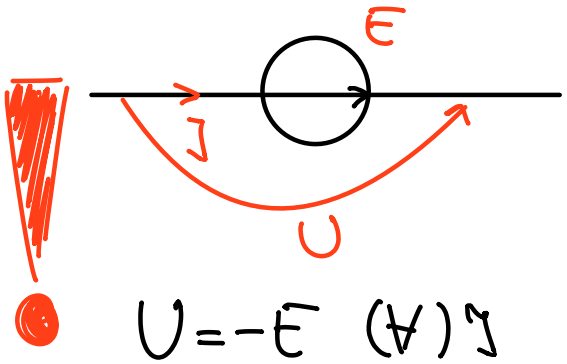
(2) Sursa ideală de tensiune



tensiunea electromotoare (V)

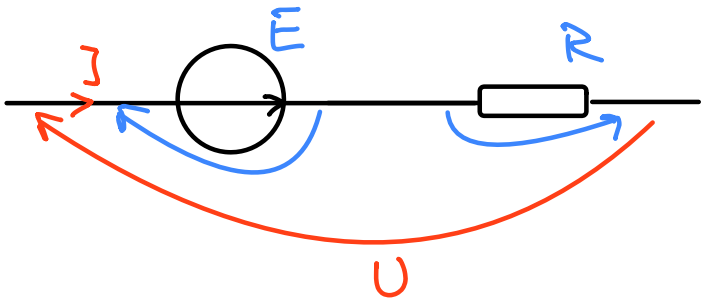
$$U = E \quad (\forall) J$$

$$P = Ui = Ei \gtrless 0$$



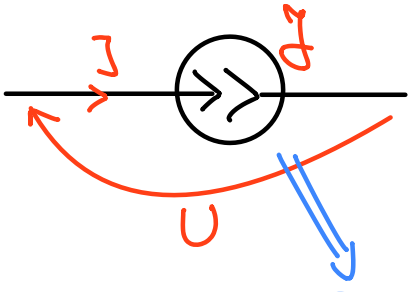
$$U = -E \quad (\forall) J$$

(3) Sursă reală de tensiune



$$U = E - JR$$

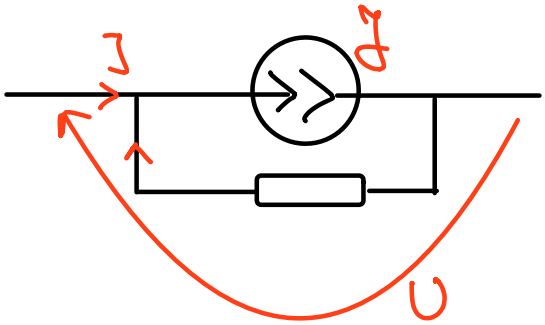
(4) Sursa ideală de curent



$$i = J \quad (\forall) U$$

$$P = Ui = UJ \gtrless 0$$

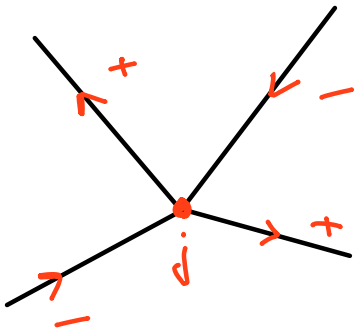
(5) Sursă reală de curent



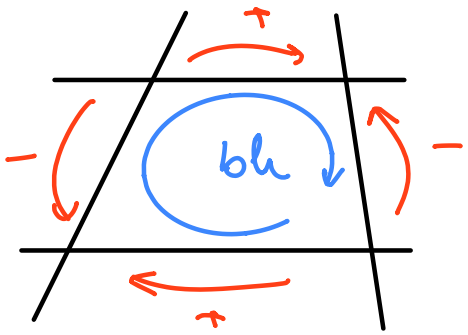
$$J = J - \frac{U}{R}$$

Teoremele lui Kirchoff

TK1 $\sum_{k \in \mathcal{N}_j}^{(A)} i_k = 0$ în nod



TK2 $\sum_{k \in b_h}^{(A)} U_k = 0$ în buclă



Bilanțul puterilor

$P_{gen} = P_{cons}$

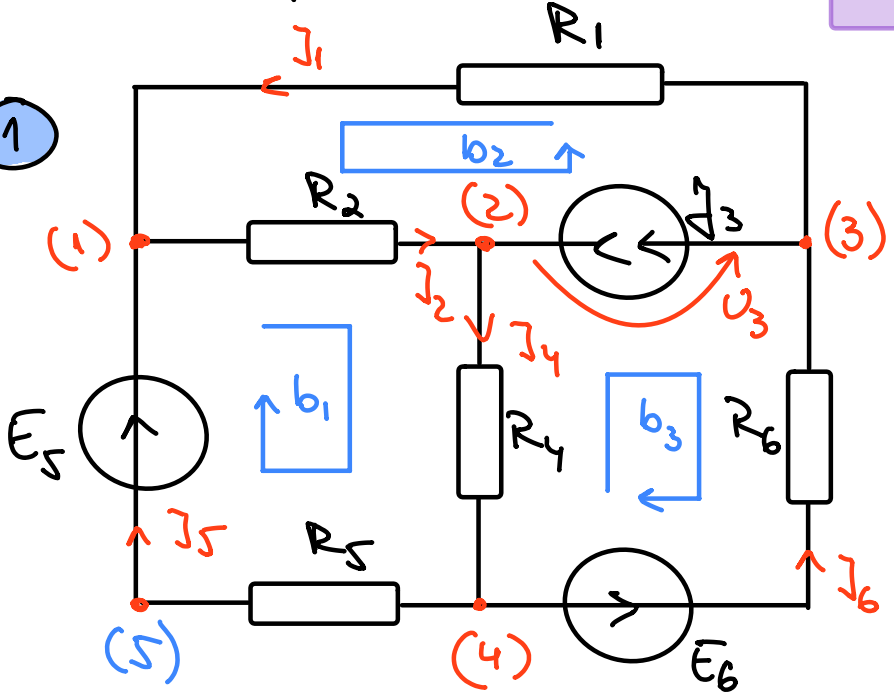
P_{gen} = putere generată

P_{cons} = putere consumată

$$P_{gen} = \sum_k (E_k i_k + U_k)$$

$$P_{cons} = \sum_k R_k i_k^2$$

1



$R_1 = 20\,\Omega$

$R_2 = 10\,\Omega$

$J_3 = 4\,A$

$E_5 = 70\,V$

$R_4 = R_5 = 5\,\Omega$

$E_6 = 30\,V$

$R_6 = 10\,V$

de luat în calcul

doar la calculator

Pas 1 noduri — 4

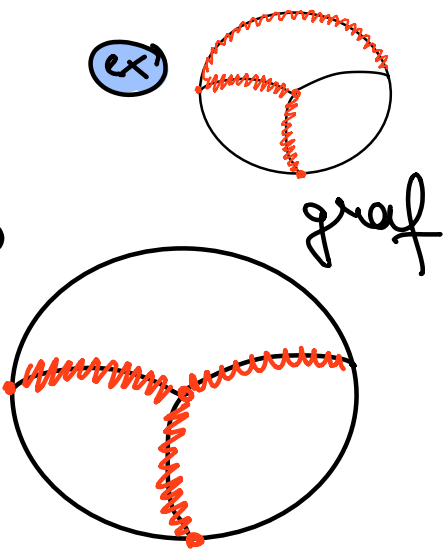
laturi — 6

bucle — 3 $(L-N+1)$

Pas 2 Sensuri arbitrare pt. i_k (desen)

Pas 3 Tk1
$$\begin{cases} -J_1+J_2-J_5=0 \\ -J_2-J_3+J_4=0 \\ J_1+J_3-J_6=0 \end{cases}$$

Pas 4 Tk2
$$\begin{cases} [b_1]: R_2J_2+R_4J_4+R_5J_5-E_5=0 \\ [b_2]: R_1J_1+R_2J_2+U_3=0 \\ [b_3]: -R_4J_4+U_3-R_6J_6+E_6=0 \end{cases}$$



Pas 5 Rezolvare sistem pt $\begin{bmatrix} J_1 \\ J_2 \\ U_3 \\ J_4 \\ J_5 \\ J_6 \end{bmatrix}$

$$\begin{cases} -J_1+J_2-J_5=0 \\ -J_2+J_4=4 \\ J_1-J_6=-4 \\ 10J_2+5J_4+5J_5=70 \\ 20J_1+10J_2+U_3=0 \\ -5J_4+U_3-10J_6=-30 \end{cases}$$

$$\begin{bmatrix} -1 & 1 & 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -1 \\ 0 & 10 & 0 & 5 & 5 & 0 \\ 20 & 10 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -5 & 0 & -10 \end{bmatrix} \begin{bmatrix} J_1 \\ J_2 \\ U_3 \\ J_4 \\ J_5 \\ J_6 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ -4 \\ 70 \\ 0 \\ -30 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 & 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -1 \\ 0 & 2 & 0 & 1 & 1 & 0 \\ 20 & 10 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -5 & 0 & -10 \end{bmatrix} \begin{bmatrix} J_1 \\ J_2 \\ U_3 \\ J_4 \\ J_5 \\ J_6 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ -4 \\ 14 \\ 0 \\ -30 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 & 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -5 & 0 & -10 \\ 1 & 0 & 0 & 0 & 0 & -1 \\ 0 & 2 & 0 & 1 & 1 & 0 \\ 20 & 10 & 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} J_1 \\ J_2 \\ U_3 \\ J_4 \\ J_5 \\ J_6 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ -30 \\ -4 \\ 14 \\ 0 \end{bmatrix}$$

$L_4 + L_1 + L_2$

$$\begin{bmatrix} -1 & 1 & 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -5 & 0 & -10 \\ 0 & 0 & 0 & 1 & -1 & -1 \\ 0 & 2 & 0 & 1 & 1 & 0 \\ 20 & 10 & 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} J_1 \\ J_2 \\ U_3 \\ J_4 \\ J_5 \\ J_6 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ -30 \\ 0 \\ 14 \\ 0 \end{bmatrix}$$

$L_5 + 2L_2 - 3L_4$

$$\begin{bmatrix} -1 & 1 & 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -5 & 0 & -10 \\ 0 & 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 4 & 3 \\ 20 & 10 & 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} J_1 \\ J_2 \\ U_3 \\ J_4 \\ J_5 \\ J_6 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ -30 \\ 0 \\ 22 \\ 0 \end{bmatrix}$$

$L_6 + 20L_4 + 30L_2 - L_3 - 35L_4 - \frac{15}{4}L_5$

$$\begin{bmatrix} -1 & 1 & 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -5 & 0 & -10 \\ 0 & 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 4 & 3 \\ 0 & 0 & 0 & 0 & 0 & 135/4 \end{bmatrix} \begin{bmatrix} J_1 \\ J_2 \\ U_3 \\ J_4 \\ J_5 \\ J_6 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ -30 \\ 0 \\ 22 \\ 135/2 \end{bmatrix} \longrightarrow$$

$$\longrightarrow \begin{cases} J_1 = -2A \\ J_2 = 2A \\ U_3 = 20V \\ J_4 = 6A \\ J_5 = 4A \\ J_6 = 2A \end{cases}$$

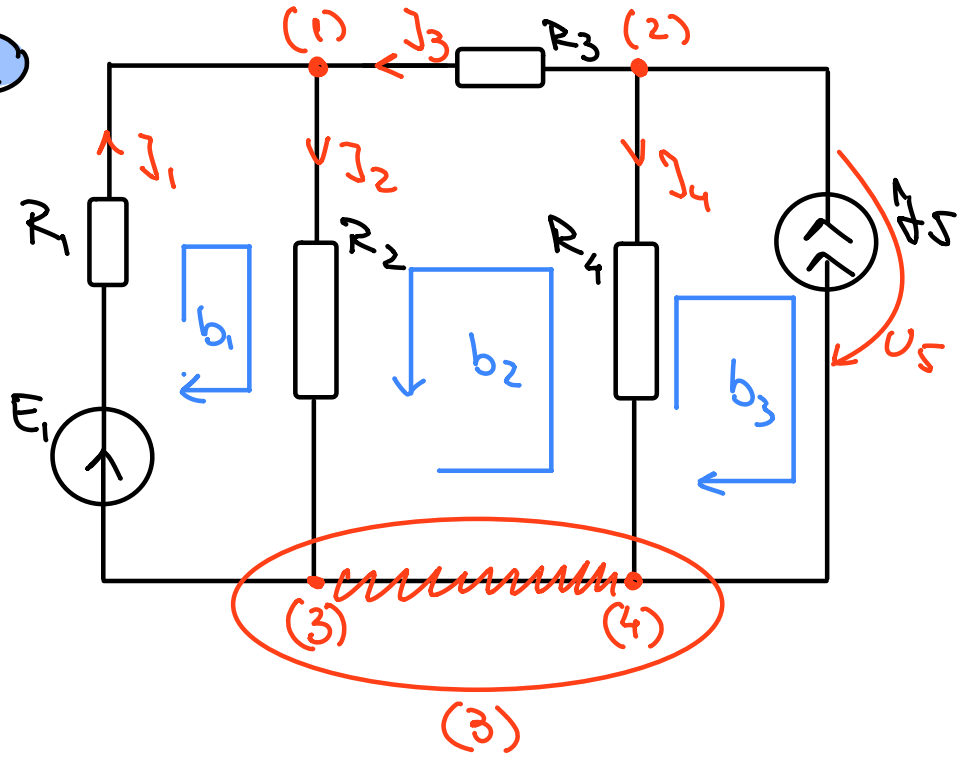
Pas 6 Bilanț de puteri

$$P_{gen} = E_5 J_5 + E_6 J_6 + U_3 J_3 = 70 \cdot 4 + 30 \cdot 2 + 20 \cdot 4 = 420\ W$$

$$P_{cons} = R_1 J_1^2 + R_2 J_2^2 + R_4 J_4^2 + R_5 J_5^2 + R_6 J_6^2$$

$$= 20 \cdot 4 + 10 \cdot 4 + 5 \cdot 36 + 5 \cdot 16 + 10 \cdot 4 = 420\ W$$

②



$R_1 = R_2 = 2\Omega$

$R_3 = R_4 = 1\Omega$

$E_1 = 12\ V$

$J_5 = 12\ A$

$N = 3$

$L = 5$

$B = 4$

Tk1: $J_1 + J_3 - J_2 = 0$

$J_5 - J_3 - J_4 = 0$

Tk2: $R_1 J_1 + R_2 J_2 - E_1 = 0$

$J_2 R_2 + J_3 R_3 - J_4 R_4 = 0$

$U_5 - J_4 R_4 = 0$

$$\left|\begin{array}{ccccc|c} 1 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & -1 & 0 & -12 \\ 2 & 2 & 0 & 0 & 0 & 12 \\ 0 & 2 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 \end{array}\right| \leftrightarrow$$

$$\left|\begin{array}{ccccc|c} 1 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 12 \\ 1 & 1 & 0 & 0 & 0 & 6 \\ 0 & 1 & 0 & -1 & 0 & -6 \\ 0 & 0 & 0 & -1 & 1 & 0 \end{array}\right| \leftrightarrow$$

$$\begin{cases} 2J_1 + J_3 = 6 \\ J_3 + J_4 = 12 \end{cases} \Rightarrow 2J_1 - J_4 = -6$$

$$\begin{cases} J_1 - J_2 = 6 \\ J_2 - J_4 = -6 \end{cases} \rightarrow \begin{array}{l} J_1 + J_4 = 12 \\ 3J_1 = 6 \end{array}$$

$$\boxed{J_1 = 2A}$$

$$\left.\begin{array}{l} J_2 = 6 - J_1 = 4A \\ J_4 = J_2 + 6 = 10A \\ J_3 = 12 - J_4 = 2A \end{array}\right| \Rightarrow -J_4 + U_5 = 0 \Rightarrow U_5 = J_4 = 10\,V$$

$$P_{gen} = E_1 J_1 + U_5 J_5 = 12 \cdot 2 + 10 \cdot 2 = 144\,W$$

$$P_{cons} = R_1 J_1^2 + R_2 J_2^2 + R_3 J_3^2 + R_4 J_4^2 = 2 \cdot 4 + 2 \cdot 16 + 1 \cdot 4 + 1 \cdot 100$$

$$P_{cons} = 144\,W$$